

MediaLedger

User Manual

Version 1.0.0

A ledger for your Plex media share — Windows desktop app and Raspberry Pi web server

AxialForge · github.com/AxialForge/medialedger

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1. What MediaLedger is

MediaLedger is a desktop application for Windows that keeps a **ledger** of everything on your media share: every TV episode, anime episode, movie file and web video, what each file is technically made of, what changed since the last look, and what is missing. It reads the files directly over the network with the ffprobe tool, stores the results in a local database on your PC, and shows the picture in a set of tabs.

It is not a player, a downloader or a replacement for Plex. It sits beside Plex and answers the questions Plex is bad at: which series are missing episodes, which movies exist in two resolutions, which shows have no subtitles, what arrived on the NAS last week, and which files carry names that Plex will struggle with.

1.1 What it never does

- It never deletes a file. Files that disappear from the share are flagged **missing** in the ledger until you press a button to forget them.
- It never writes to the share unless you turn on one of the two rename tools, tick specific files, and confirm. Everything else is read-only.
- It never sends your library anywhere. The only network calls are: the update check against GitHub, the optional ffmpeg download, episode-count lookups to TVmaze and AniList (series titles only), and your own Plex server on the LAN.
- The desktop app has no web server and no browser tab; it is a plain window. The optional Raspberry Pi build (chapter 22) is the same program served to your LAN behind a password.
- It never needs an account, a subscription or an API key.

1.2 Core concepts

Term	Meaning
Root	A folder MediaLedger watches, with a type: TV, Anime, Movies, Web videos or Adult. Usually a UNC path such as \\192.168.1.204\Apocrypha_Media_Pool\Anime.
Scan	One pass over every enabled root: list files, compare with the ledger, parse names, probe new or changed files, write the change log, export CSVs.
Parse	Working out series, season, episode, title and year from the folder and file name. Fast and offline.
Probe	Running ffprobe on a file to read its real resolution, length, codecs, languages and subtitle tracks. This is the slow part; it only happens for new or changed files.
Fix	A manual correction you make to a parsed result. Fixes live in the database and are re-applied on every scan, so a corrected file stays corrected.
Change log	The record of what each scan found: added, removed, modified and returned files, plus errors.
Expected episodes	Per-season episode counts fetched from TVmaze (TV) or AniList (anime), used to say exactly which episodes you lack.
Placeholder	A literal word such as Year or Source that the movie naming engine writes into a name when it cannot prove the value. Deliberately visible so you can fill it in later.

2. Installing and first run

2.1 Install

1. Download `medialedger-<version>-setup.exe` from the Releases page of the repository.
2. Run it. It is a one-click, per-user install with no administrator prompt. The build is not code-signed, so Windows SmartScreen will show a blue box the first time: click **More info**, then **Run anyway**.
3. The app opens automatically when the installer finishes and is added to the Start menu.

2.2 First launch

On the first launch MediaLedger checks for `ffmpeg`. If an `ffmpeg` install exists under `C:\ffmpeg`, via `winget` or `scoop`, or on your `PATH`, it is used. If nothing is found, the app downloads the latest static Windows build of `ffmpeg` from the BtbN project into its own data folder and points itself at it. A toast in the corner shows progress; it is about 100 MB.

Next, open **Settings** → **Library roots** and point each root at your shares. The defaults match the AxialForge NAS layout; change the paths if yours differ. Press **Save settings**, then **Scan now** in the bottom-left corner.

The first scan probes every file and takes a while: about 25 minutes for 24,000 files over gigabit SMB. Every later scan only probes new or changed files and finishes in seconds. You can close the window during a scan started by the scheduler, but a manual scan is cancelled if you quit.

2.3 Where your data lives

Item	Location
Database, settings, log, downloaded <code>ffmpeg</code>	<code>%APPDATA%\MediaLedger\</code>
Automatic database backups	<code>%APPDATA%\MediaLedger\medialedger.db.backups\</code> (taken before every schema upgrade; newest 10 kept)
CSV exports	<code>%APPDATA%\MediaLedger\exports\<timestamp>\</code> plus a latest\ copy
The program itself	<code>%LOCALAPPDATA%\Programs\MediaLedger\</code>

Because the data sits in your user profile and not in the program folder, reinstalling or updating never loses the ledger, your fixes or your ratings.

2.4 Updates

The installed app checks GitHub Releases on launch and every six hours, downloads a newer version silently and installs it the next time you close the app. The About page has a **Check for updates** button and shows the current state. No account or token is needed.

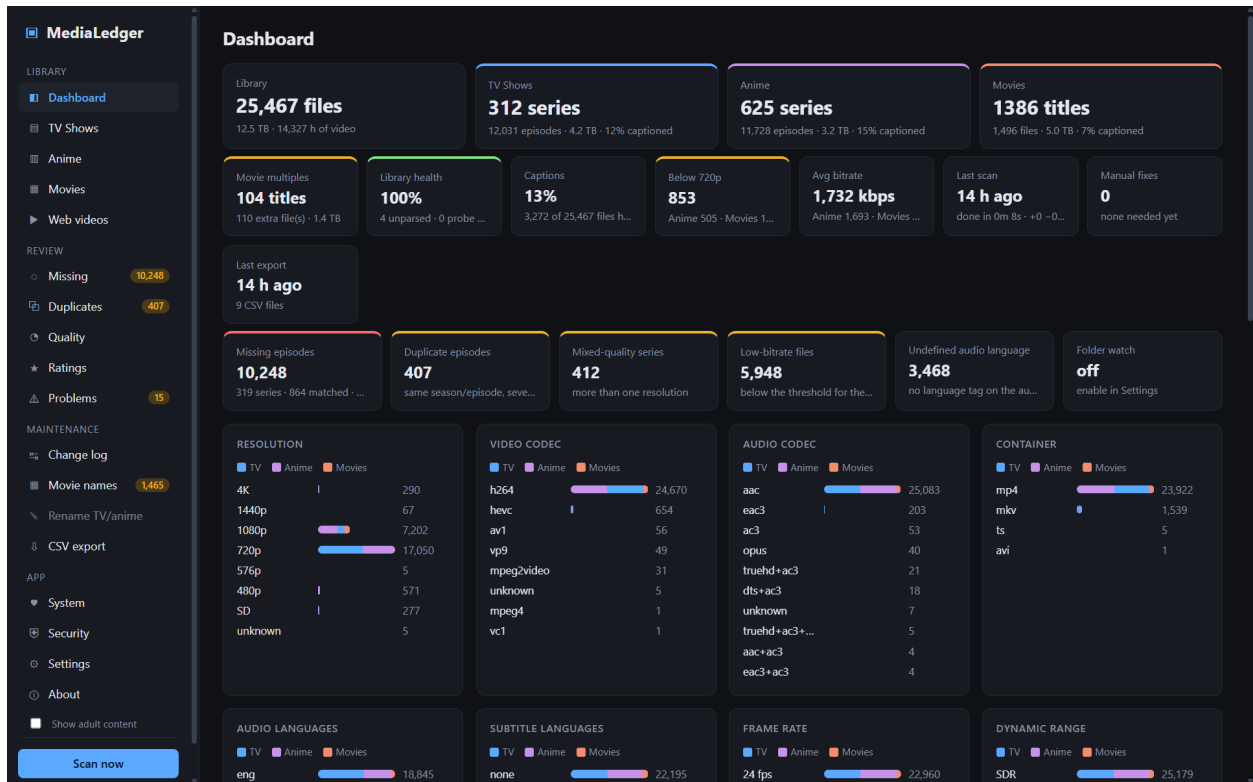
3. The window

The left sidebar is the only navigation. It is grouped into **Library** (what you have), **Review** (what needs attention), **Maintenance** (tools that change or export things) and **App**. Orange counters on Missing, Duplicates, Problems and Movie names show how many items wait for you.

- **Scan now** starts a manual scan. While a scan runs the button becomes **Cancel scan** and a progress bar shows the phase (listing, indexing, probing), the rate and an estimate of the time left.
- A second thin progress bar appears while expected-episode lookups or a Plex sync run in the background.
- The **Show adult content** switch appears only when an Adult root is configured. It is off on every launch.
- The version in the bottom-left corner shows "(dev)" when the app is run from source rather than installed.

Every table in the app can be sorted by clicking a column heading (click again to reverse) and filtered with the search box above it. Many rows are clickable: a series opens its episodes, a movie title opens its files, an episode or movie file reveals itself in Windows Explorer.

4. Dashboard



The Dashboard after a full scan.

The Dashboard is a summary of the whole ledger. Everything on it is computed live from the database.

4.1 Top tiles

Tile	What it shows
Library	Total files, total size and total hours of video across every non-adult root.
TV Shows / Anime / Movies	Number of series or titles, number of files, size, and the share of files that have captions (embedded or sidecar subtitles).
Movie multiples	Titles that exist as more than one file, how many extra files that is, and their combined size.
Library health	100% minus the share of files that are unparsed, failed to probe, or are missing.
Captions	Files with any subtitle track or sidecar.
Below 720p	Files whose probed resolution is 576p, 480p or SD.
Avg bitrate	Average video bitrate, overall and per library.
Last scan	When it ran, how long it took, and its added / removed / modified counts.
Manual fixes	How many corrections you have saved.
Last export	When CSVs were last written and how many files.
Missing episodes	Episodes you lack according to TVmaze / AniList, how many series have gaps, and how many series are matched, unmatched or still pending.

Duplicate episodes	Season/episode combinations that exist as several files.
Mixed-quality series	Series with more than one resolution.
Low-bitrate files	Files under the bitrate threshold for their resolution (thresholds are in Settings).
Undefined audio language	Files whose audio track carries no language tag.
Folder watch	Whether the optional folder watcher is on and when it last saw a change.

4.2 Charts

Eight stacked bar charts break the probed files down by resolution, video codec, audio codec, container, audio language, subtitle language, frame rate and dynamic range. Each bar is split by library (blue TV, purple Anime, orange Movies); hover a segment for the exact count.

4.3 Panels

- **Recently added** – the newest files by first-seen time, with a link to the change log.
- **Largest series** and **Largest movie files** – by size on disk.
- **Most missing episodes** – the series with the biggest gaps, with the exact episodes listed; whole missing seasons collapse into ranges such as "S6–S12 entirely".
- **Scan history** – the last eight scans with status, trigger, thread count, duration and counts.

5. TV Shows and Anime

MediaLedger

Anime

Series: **625** | Episodes: **11,728** | Size: **3.2 TB** | Runtime: **4,664 h** | Full captions: **84 series** | With issues: **1 series**

Filter... 625 of 625

SERIES	SEASONS	EPISODES	RUNTIME	SIZE	RESOLUTION	CODEC	AUDIO	SUBS
4 Cut Hero	1	10	1.3 h	406 MB	720p	h264	und	
7th Time Loop The Villainess Enjoys a Carefree Life Married to Her Worst Enemy!	1	12	4.8 h	1.2 GB	720p 1080p	h264	jpn	
365 Days to the Wedding	1	12	4.7 h	4.2 GB	1080p 720p	h264	jpn eng	
A Centaur's Life	1	12	4.3 h	1.3 GB	720p 1080p	h264	eng und	
A Certain Magical Index	3 S1-S3	75	31 h	45.4 GB	1080p	h264	eng jpn	
A Certain Middle-Aged Man's VRMMO Activity Log	1	12	4.7 h	1.2 GB	720p	h264	jpn	
A Condition Called Love	1	12	4.7 h	1.7 GB	1080p 720p	h264	eng	
A Couple of Cuckoos	2 S1-S2	32	12 h	10.5 GB	720p	h264	und	
A Galaxy Next Door	1	12	4.8 h	1.0 GB	720p 1080p	h264	eng	
A Gatherer's Adventure in Isekai	1	12	4.7 h	2.5 GB	720p	h264	eng eng	
A Journey Through Another World Raising Kids While Adventuring	1	12	4.8 h	2.2 GB	720p 1080p	h264	jpn	
A Livid Lady's Guide to Getting Even How I Crushed My Homeland with My Mighty Grimoires	1	7	2.8 h	2.5 GB	720p	h264	jpn eng	
A Misanthrope Teaches a Class for Demi-Humans	1	1	0.4 h	324 MB	720p	h264	eng eng	
A Nobody's Way Up to an Exploration Hero	1	12	4.7 h	2.6 GB	1080p	h264	jpn	
A Place Further Than the Universe	1	13	5.1 h	13.9 GB	1080p	h264	jpn eng	

The Anime list. TV Shows looks the same.

One row per series. The summary strip counts series, episodes, size, runtime, how many series have full captions and how many have parse issues.

Column	Meaning
Seasons	Distinct season numbers on disk, with the range when it is not 1..N.
Episodes	Files that belong to the series.
Runtime / Size	Totals from the probe.
Resolution / Codec	Every resolution and codec present.
Audio / Subs	Union of audio and subtitle languages across the series.
Rating	The online average (TVmaze or AniList) once the series has been matched.
Mine	Your own 0–5 stars. Click a star to set it; click the same star again to clear. A small "P4.5" beside it is your Plex rating when Plex is synced.
Watched	Share of episodes Plex has marked played (needs a Plex sync).
Captions	Share of episodes with subtitles: green 100%, amber partial, red none.
Missing	Episodes you lack out of the expected total, or "complete", or "no match" when the online lookup could not find the series.
Issues	Unparsed or unprobed files in the series.

5.1 Episode detail

The screenshot shows the MediaLedger interface for the 'One Piece' anime series. The left sidebar contains navigation options like Dashboard, TV Shows, Anime, Movies, Web videos, Review, Missing (10,248), Duplicates (407), Quality, Ratings, Problems (15), Change log, Movie names (1,465), Rename TV/anime, CSV export, APP, System, Security, Settings, and About. The main content area shows the 'One Piece' series with 215 episodes, 57.9 GB, and 85 h. The 'Expected Episodes' section shows 1176 episodes, with 1115 missing and 61 present. The 'Match...' button is visible. Below this, a table lists episodes with columns: EP, TITLE, FILE, LENGTH, RES, FPS, VIDEO, AUDIO, SUBS, CAPTIONS, KBPS, SIZE, and Fix... The table shows episodes S01E01 through S01E10.

EP	TITLE	FILE	LENGTH	RES	FPS	VIDEO	AUDIO	SUBS	CAPTIONS	KBPS	SIZE	Fix...
S01E01	I'm Luffy! The Man Who's Gonna Be King of the Pirates!	One Piece_S01E01_I'm Luffy! The Man Who's Gonna Be King of the Pirates!.mp4	24m 57s	480p	23.976	h264	und aac stereo	no	1963	350 MB	Fix...	
S01E02	Enter the Great Swordsman! Pirate Hunter Roronoa Zoro!	One Piece_S01E02_Enter the Great Swordsman! Pirate Hunter Roronoa Zoro!.mp4	24m 57s	480p	23.976	h264	und aac stereo	no	2025	362 MB	Fix...	
S01E03	Morgan versus Luffy! Who's the Mysterious Pretty Girl?	One Piece_S01E03_Morgan versus Luffy! Who's the Mysterious Pretty Girl!.mp4	24m 57s	480p	23.976	h264	und aac stereo	no	1981	353 MB	Fix...	
S01E04	Luffy's Past! Enter Red-Haired Shanks!	One Piece_S01E04_Luffy's Past! Enter Red-Haired Shanks!.mp4	24m 57s	480p	23.976	h264	und aac stereo	no	1968	351 MB	Fix...	
S01E05	A Terrifying Mysterious Power! Captain Buggy, the Clown Pirate!	One Piece_S01E05_A Terrifying Mysterious Power! Captain Buggy, the Clown Pirate!.mp4	24m 57s	480p	23.976	h264	und aac stereo	no	1981	354 MB	Fix...	
S01E06	Desperate Situation! Beast Tamer Mohji Vs Luffy!	One Piece_S01E06_Desperate Situation! Beast Tamer Mohji Vs. Luffy!.mp4	24m 57s	480p	23.976	h264	und aac stereo	no	1961	350 MB	Fix...	
S01E07	Epic Showdown! Swordsman Zoro Vs Acrobat Cabaji!	One Piece_S01E07_Epic Showdown! Swordsman Zoro Vs. Acrobat Cabaji!.mp4	24m 57s	480p	23.976	h264	und aac stereo	no	2074	370 MB	Fix...	
S01E08	Who Is the Victor Devil Fruit Power Showdown!	One Piece_S01E08_Who Is the Victor_ Devil Fruit Power Showdown!.mp4	24m 57s	480p	23.976	h264	und aac stereo	no	2074	370 MB	Fix...	
S01E09	The Honorable Liar Captain Usopp!	One Piece_S01E09_The Honorable Liar_ Captain Usopp!.mp4	24m 57s	480p	23.976	h264	und aac stereo	no	2006	358 MB	Fix...	
S01E10	The Weirdest Guy Ever! Jango the Hypnotist!	One Piece_S01E10_The Weirdest Guy Ever! Jango the Hypnotist!.mp4	24m 57s	480p	23.976	h264	und aac stereo	no	1965	351 MB	Fix...	

A series opened from the Anime list, with the expected-episode grid at the top.

Click a series to open it. The header shows the totals and a **Match...** button (see chapter 10). When expected counts exist, a grid shows every season with each episode number as a green (present) or red (missing) cell.

The table lists every file with its parsed episode number, title, file name, probed length, resolution and pixel size, fps, video codec with bit depth and HDR badge, audio languages and codecs, subtitle tracks, captions flag, bitrate and size. Badges mark files that are **missing** from disk or have a manual **fixed** override. Click a row to reveal the file in Explorer; click **Fix...** to correct its details (chapter 14.2).

6. Movies

MediaLedger

LIBRARY

- Dashboard
- TV Shows
- Anime
- Movies**
- Web videos

REVIEW

- Missing **10,248**
- Duplicates **407**
- Quality
- Ratings
- Problems **15**

MAINTENANCE

- Change log
- Movie names **1,465**
- Rename TV/anime
- CSV export

APP

- System
- Security
- Settings
- About
- Show adult content

Scan now

Movies

Titles: **1386** | Files: **1496** | Size: **5.0 TB** | Multiples: **104 titles** (1.4 TB) | With captions: **95**

Filter... | 1,386 of 1,386 | ☐ Only titles with multiple files

TITLE	YEAR	FILES	VERSIONS	LENGTH	CODEC	AUDIO	SUBS
6 Underground	2019	1	1080p	2h 8m	h264	und	
12 Strong	2018	1	1080p	2h 9m	h264	und	
13 Hours The Secret Soldiers of Benghazi	2016	1	1080p	2h 24m	h264	und	
21 Jump Street	2012	1	1080p	1h 49m	h264	und	
22 Jump Street x2	2014	2	1080p 1080p	1h 51m	hevc h264	eng	
28 Days	2000	1	1080p	1h 43m	h264	und	
47 Ronin	2013	1	1080p	1h 58m	h264	eng	
57 Seconds	2023	1	1080p 1080p	1h 39m	h264	eng	
65	2023	1	1080p	1h 32m	h264	eng	
101 Dalmatians	1961	1	1080p	1h 19m	h264	und	
101 Dalmatians 2 Patches London Adventure	2002	1	720p 720p	1h 13m	h264	und	
222	2017	1	1080p 1080p	1h 38m	h264	und	
300	2007	1	1080p	1h 56m	h264	eng	
1917	2019	1	1080p	1h 58m	h264	eng	
1992	2022	1	4K 4K	1h 37m	h264	eng	
2001 A Space Odyssey	1968	1	1080p 1080p	2h 28m	h264	eng	
2012	2009	1	1080p	2h 37m	h264	eng	
A Breed Apart x2	2025	2	1080p 4K	1h 39m	h264	eng	
A Bronx Tale	1993	1	1080p 1080p	2h 1m	h264	eng	
A Charlie Brown Christmas	1966	1	1080p 1080p	1h 12m	h264	und	

The Movies list; a x2 badge marks titles with two files.

One row per title. Files are grouped by a normalised title plus year, so Pacific Rim (2013).mp4 and Pacific Rim (2013) [4k].mkv become one title with two versions. The **Only titles with multiple files** checkbox narrows the list to those.

The screenshot shows the MediaLedger application interface. On the left is a dark sidebar with a 'MediaLedger' header and a menu. The menu is divided into sections: LIBRARY (Dashboard, TV Shows, Anime, Movies, Web videos), REVIEW (Missing: 10,248, Duplicates: 407, Quality, Ratings, Problems: 15), MAINTENANCE (Change log, Movie names: 1,465, Rename TV/anime, CSV export), and APP (System, Security, Settings, About, Show adult content). A 'Scan now' button is at the bottom of the sidebar. The main content area is titled 'Pacific Rim (2013) 2 file(s)'. It contains a table with the following data:

FILE	EDITION / TAG	LENGTH	RES	FPS	VIDEO	AUDIO	SUBS	CAPTIONS	KBPS
Pacific Rim (2013) [2160p HDR Web Remux].mkv	2160p HDR Web Remux	2h 11m	4K 3840x2160	23.976	hevc Main 10 10-bit HDR10	hixeng eac3aac 5.1(side);7.1	eng 1 track(s)	yes	7758
Pacific Rim (2013).mp4		2h 11m	1080p 1920x1080	23.976	h264 High	eng aac stereo		no	2125

The files behind one title.

Click a title to see its files side by side: edition tag, length, resolution, fps, video codec and profile, audio, subtitles, captions, bitrate, size and container. Each row has **Fix...** for corrections and reveals the file in Explorer when clicked.

7. Web videos

Web videos

Downloaded web videos grouped by channel folder. Titles come from the file name with yt-dlp ids and dates stripped when present.

Channels: **10** Videos: **212** Size: **48.4 GB** Runtime: **44 h**

Filter... 10 of 10

CHANNEL	VIDEOS	RUNTIME	SIZE	RESOLUTION	UPLOADS	LAST ADDED
(no channel)	14	5.0 h	4.4 GB	1080p 4K	—	16 h ago
AFVK	108	12 h	8.1 GB	720p 1440p 1080p	—	16 h ago
Demos	3	0.3 h	1.9 GB	4K 1080p	—	16 h ago
Kurzgesagt – In a Nutshell	28	9.0 h	8.8 GB	4K 1080p 720p	—	16 h ago
Melody Sheep	11	5.5 h	14.7 GB	4K 1440p 1080p	—	16 h ago
Movie & Tv Trailers	20	0.8 h	2.2 GB	4K 1080p 1440p	—	16 h ago
Music Videos	2	0.1 h	82.8 MB	1080p	—	16 h ago
Short Films	5	0.5 h	553 MB	1080p 4K 1440p	—	16 h ago
Smosh	19	9.6 h	7.3 GB	1080p	—	16 h ago
The Fat Electrification	2	0.9 h	521 MB	1080p	—	16 h ago

Web videos grouped by channel folder.

A root of type **Web videos** is for downloaded web content, typically from yt-dlp. The top-level folder is treated as the channel; loose files at the root fall under "(no channel)". The file name becomes the title after stripping a trailing [videoId] and any YYYYMMDD date. When an id is present the video list links to it on YouTube.

The channel list shows video count, runtime, size, resolutions, the upload-date range and when the last file arrived. Open a channel to see its videos with every probed field. A channel can be rated with stars from its header.

8. Adult library

A root of type **Adult** holds content you want kept apart. MediaLedger classifies each file under it as anime, TV or movie from its folder and name, so the ordinary series and movie views and tools all work on it. What differs is visibility:

- Adult files are excluded from every list, count, chart and query until the **Show adult content** switch at the bottom of the sidebar is on.
- The switch is **off on every launch**. Nothing remembers it.
- While it is on, an **Adult** tab appears under Library with its own summary, resolution chart, recently added, series list, movie list and unparsed list. Series open in the normal episode view.
- Adult files are left out of CSV exports unless **Settings → Adult content → Include in CSV exports** is ticked.
- Screenshots and the README never include adult content.

Under Settings → Adult content you can also choose whether files that match none of the classification patterns are treated as anime or TV.

9. Missing episodes

Missing episodes

Expected episode counts come from TVmaze (TV) and AniList (anime), fetched in the background after each scan. Compared with what is on disk per season. Use **Match...** when a series was matched to the wrong entry, was not found, or you want to enter counts by hand.

Series with gaps: **319** (10,248 episodes missing)

Complete series: **545**

No match found: **72** (use Match... to search)

Not looked up yet: **0**

Locked by you: **0** (manual matches or counts)

Look up new series | Re-check all unlocked series | Rate-limited: roughly 1-2 series per second.

Filter... 936 of 936

LIBRARY	SERIES	SOURCE	STATUS	EXPECTED	HAVE	MISSING	WHICH	
Anime	One Piece matched: ONE PIECE	anilist	RELEASING	1176	61	1115	S1: 62,63,64,65,66,67,68,69,... (1115)	Match...
TV	The Simpsons	tvnmaze	Running	802	90	712	S5: 10,11,12,13,14,15,16,17,... (13); S6-S38 entirely	Match...
TV	Diners, Drive-ins and Dives matched: Diners, Drive-Ins and Dives	tvnmaze	Running	568	0	568	S2007-S2026 entirely	Match...
TV	Thomas & Friends	tvnmaze	Ended	515	156	359	S7-S21 entirely	Match...
TV	Murdoch Mysteries	tvnmaze	Running	330	27	303	S1: 2,3,4,5,6,7,8,9,... (12); S2-S3, S6-S19 entirely	Match...
TV	How It's Made	tvnmaze	Ended	417	128	289	S2: 6; S4: 3; S10: 14; S11-S32 entirely	Match...
TV	Bob the Builder	tvnmaze	Ended	250	13	237	S2-S18 entirely	Match...
TV	King of the Hill	tvnmaze	Running	279	58	221	S3: 24,25; S4-S15 entirely	Match...
Anime	Bleach	anilist	RELEASING	416	195	221	S1: 196,197,198,199,200,201,202,203,... (171); S2-S5 entirely	Match...
TV	Regular Show	tvnmaze	Ended	246	40	206	S3-S8 entirely	Match...
TV	South Park	tvnmaze	Running	335	153	182	S11-S23, S25-S29 entirely	Match...
TV	The Walking Dead	tvnmaze	Ended	177	6	171	S2-S11 entirely	Match...
TV	Castle	tvnmaze	Ended	173	10	163	S2-S8 entirely	Match...
TV	Modern Family	tvnmaze	Ended	250	120	130	S6-S11 entirely	Match...

S1: 27,28,29,30,31,32,33,34,... (21); S2: 40,41,42,43,44,45,46,47,... (25); S3: ...

Series ranked by how many episodes are missing.

After each scan MediaLedger looks up every series it has not seen before on TVmaze (TV) or AniList (anime) and stores the number of episodes in each season. Both services are free and need no key; lookups run at about one series per second in the background, and airing series are re-checked every two weeks. Comparing those counts with what is on disk, per season, gives an exact list of what you lack.

Column	Meaning
Source	tvnmaze, anilist, manual or none. A lock badge means you chose the match or entered counts yourself; automatic re-checks never overwrite a locked series.
Status	Running / Ended (TVmaze) or RELEASING / FINISHED (AniList).
Expected / Have / Missing	Regular-season episodes only; specials (season 0) are not counted.
Which	The missing episodes per season. Whole missing seasons collapse into ranges.
absolute numbering	A badge shown when a season on disk is numbered far beyond its expected length (for example One Piece folders numbered 62–77). That season is skipped rather than reported as missing.

9.1 The Match dialog

Press **Match...** on any series (here, on the series page, or in the lists) when a series was matched to the wrong entry, was not found, or you know the counts yourself. The dialog shows the current match, lets you search either TVmaze or AniList regardless of library type, and lists candidates with year, format and

episode count; click one to use it. Below that you can type counts per season by hand, declare that the series has no expected counts, or return a locked series to automatic.

AniList numbers anime by cour, so a split season shows up as separate "Part 2" entries. MediaLedger merges those into the same season automatically. If a series still lines up better with broadcast seasons, switch its source to TVmaze in the Match dialog.

The two buttons above the list start a background pass: **Look up new series** for anything not yet looked up, **Re-check all unlocked series** to refresh everything that you have not locked.

10. Duplicates

MediaLedger

LIBRARY

- Dashboard
- TV Shows
- Anime
- Movies
- Web videos

REVIEW

- Missing **10,248**
- Duplicates 407**
- Quality
- Ratings
- Problems **15**

MAINTENANCE

- Change log
- Movie names **1,465**
- Rename TV/anime
- CSV export

APP

- System
- Security
- Settings
- About
- Show adult content

Scan now

Duplicate episodes

Episodes that exist as more than one file. Compare them side by side and mark the one to **keep**; the others are flagged as discard candidates. MediaLedger never deletes anything — use **Reveal** to open the file in Explorer and decide there. Decisions are remembered across scans.

Duplicate episodes **407**

Extra files **427**

Space in extras **131 GB**
size of all but the largest file per episode

Decided **0**
407 still to review

Filter by series... ☐ Hide decided 300 of 407

TV Altered Carbon S01E01 2 files

Altered Carbon S01E01 - 720p WEB-DL.mp4 **BEST QUALITY**

Size **451 MB** Length **58m 54s**
Res **720p** 1280×640 Bitrate **1,070 kbps**
Video **h264** Audio **eng aac**
Subs **0** Type **mp4**

1 Altered Carbon Season 1.mkv

Size **332 MB** Length **59m 4s**
Res **720p** 1280×640 Bitrate **785 kbps**
Video **hevc** 10-bit Audio **hin;eng aac**
Subs **5 eng;gerspa;fre;dut** Type **mkv**

Keep this **Reveal** **Fix...**

TV Altered Carbon S01E02 2 files

Altered Carbon S01E02 - 720p WEB-DL.mp4 **BEST QUALITY**

Size **403 MB** Length **56m 52s**
Res **720p** 1280×640 Bitrate **960 kbps**
Video **h264** Audio **eng aac**
Subs **0** Type **mp4**

2 Altered Carbon Season 1.mkv

Size **352 MB** Length **57m 2s**
Res **720p** 1280×640 Bitrate **863 kbps**
Video **hevc** 10-bit Audio **hin;eng aac**
Subs **5 eng;gerspa;fre;dut** Type **mkv**

Keep this **Reveal** **Fix...**

TV Altered Carbon S01E03 2 files

Altered Carbon S01E03 - 720p WEB-DL.mp4 **BEST QUALITY**

Size **352 MB** Length **51m 7s**
Res **720p** 1280×640 Bitrate **963 kbps**
Video **h264** Audio **eng aac**
Subs **0** Type **mp4**

3 Altered Carbon Season 1.mkv

Size **263 MB** Length **51m 18s**
Res **720p** 1280×640 Bitrate **716 kbps**
Video **hevc** 10-bit Audio **hin;eng aac**
Subs **5 eng;gerspa;fre;dut** Type **mkv**

Keep this **Reveal** **Fix...**

Two files for the same episode, side by side.

Every season/episode that exists as more than one file is shown as a group, each file as a card with size, length, resolution, bitrate, codec, HDR, audio, subtitles and container. The card with the highest resolution and bitrate is marked **best quality**.

Press **Keep this** on the file you want. It is marked keep, the others are marked discard candidates and drawn faded, and the group moves to the decided state. Nothing is deleted: use **Reveal** to open a file in Explorer and delete it there if you wish. Decisions are stored with the file and survive scans and renames; **Clear decision** undoes them. **Hide decided** narrows the list to groups still waiting.

11. Quality

Quality

Files and series whose technical quality looks off: seasons that mix resolutions, files whose bitrate is unusually low for their resolution, and files with no audio, no language tag or a suspiciously short runtime. Thresholds (kbps): 4K < 6000 · 1440p < 3000 · 1080p < 1500 · 720p < 700 · 576p < 400 · 480p < 350 · SD < 250 — change them in Settings.

Mixed-resolution series
412

Mixed seasons
470
one season, several resoluti...

Low-bitrate files
5,948

Undefined audio language
3,468
Anime 2235 · Movies 384 · ...

No audio track
7

Under 2 minutes
69
samples, trailers, broken files

MIXED-RESOLUTION SERIES (412)

LIBRARY	SERIES	FILES	RESOLUTIONS	CODECS
TV	MythBusters	242	SD · 480p · 1080p · 720p	h264
TV	Futurama	199	480p · 720p · 1080p · 4K	h264
Anime	The Seven Deadly Sins	97	1080p · 480p · 720p · SD	h264
Anime	World Trigger	70	1080p · 720p · SD · 480p	h264
Anime	The Irregular at Magic High School	51	720p · SD · 1080p · 480p	h264
Anime	Astra Lost in Space	24	720p · SD · 480p · 1080p	h264
TV	Bobs Burgers	313	1080p · 480p · 720p	h264
Anime	One Piece	215	480p · 720p · 1080p	h264
Anime	My Hero Academia	160	720p · 1080p · 480p	h264 hevc
Anime	Fullmetal Alchemist Brotherhood (2009)	128	1080p · 720p · 480p	hevc h264
Anime	Food Wars	98	1080p · 720p · 480p	h264
Anime	Attack On Titan	89	1080p · 480p · 720p	h264
Anime	Kuroko's Basketball	76	720p · 1080p · 480p	h264

MIXED SEASONS (470)

LIBRARY	SERIES	SEASON	FILES	RESOLUTIONS
Anime	365 Days to the Wedding	S1	12	1080p · 720p
Anime	7th Time Loop The Villainess Enjoys a Carefree Life Married to Her Worst Enemy!	S1	12	720p · 1080p
TV	9-1-1 Lone Star	S3	18	720p · 1080p
Anime	A Centaur's Life	S1	12	720p · 1080p

The Quality report.

Lists files and series whose technical quality looks off:

- **Mixed-resolution series** and **Mixed seasons** – a series, or a single season, that holds more than one resolution.
- **Low-bitrate files** – files whose video bitrate is below the threshold for their resolution. Thresholds are editable under Settings → Quality thresholds; the defaults are 6000 kbps for 4K, 1500 for 1080p and 700 for 720p.
- **No audio track**, **Undefined audio language** and **Under 2 minutes** – usually samples, trailers, or damaged files. Use **Fix...** to ignore a file that is not real media.

12. Ratings

MediaLedger

Ratings

Online averages come from TVmaze (TV) and AniList (anime) with the episode-count lookup; nothing extra is fetched. Your own rating is a 0–5 star score per title, stored locally and exported in the CSVs. Click a star to rate; click the same star again to clear. "Plex" is the audience score Plex shows; "Plex mine" is the rating you set in Plex (shown out of 5). Watched comes from Plex play counts. Run a Plex sync from Settings to refresh them.

Titles: **2,333**

With online rating: **287** (avg 7.5 / 10)

Rated by you: **0**

Top online: **Planet Earth III** (9.3 / 10)

all libraries ☐ Only rated by me ☐ Only with a Plex rating of mine ☐ Only unrated by me

Filter... 2,333 of 2,333

LIBRARY	TITLE	ONLINE	PLEX	PLEX MINE	WATCHED	MINE	NOTE	FILES	SIZE
TV	Planet Earth III	9.3 / 10 tvmaze				★★★★★	note...	8	23.6 GB
TV	Breaking Bad	9.2 / 10 tvmaze				★★★★★	note...	62	21.4 GB
TV	The Blue Planet	9.1 / 10 tvmaze				★★★★★	note...	8	9.2 GB
TV	Chernobyl	8.9 / 10 tvmaze				★★★★★	note...	5	6.3 GB
TV	Game of Thrones	8.9 / 10 tvmaze				★★★★★	note...	73	46.4 GB
TV	Planet Earth II	8.9 / 10 tvmaze				★★★★★	note...	6	5.4 GB
TV	Gravity Falls	8.8 / 10 tvmaze				★★★★★	note...	40	6.0 GB
TV	Person of Interest	8.8 / 10 tvmaze				★★★★★	note...	7	2.8 GB
TV	Rick and Morty	8.7 / 10 tvmaze				★★★★★	note...	91	17.4 GB
TV	Star Trek The Next Generation	8.7 / 10 tvmaze				★★★★★	note...	176	54.0 GB
TV	The Expanse	8.7 / 10 tvmaze				★★★★★	note...	35	11.8 GB
TV	Better Call Saul	8.6 / 10 tvmaze				★★★★★	note...	26	10.0 GB
TV	Farscape	8.6 / 10 tvmaze				★★★★★	note...	88	32.8 GB

Online averages beside your own stars.

Column	Meaning
Online	The average score from TVmaze or AniList, out of 10, stored when the series was matched. Movies have no online score.
Plex	The audience score Plex shows for the title, once Plex is synced.
Plex mine	The rating you gave the title inside Plex, shown out of 5.
Watched	From Plex play counts: a percentage for series, yes/no for movies.
Mine	Your MediaLedger rating, 0–5 stars. Click to set, click the same star to clear.
Note	Free text saved when you leave the field.

Filters narrow the list to one library, to titles you have rated, to titles you have not, or to titles with a Plex rating of yours. Your stars and notes are exported in the series and movie CSVs.

13. Problems

MediaLedger

LIBRARY

- Dashboard
- TV Shows
- Anime
- Movies
- Web videos

REVIEW

- Missing (10,248)
- Duplicates (407)
- Quality
- Ratings
- Problems (15)**

MAINTENANCE

- Change log
- Movie names (1,465)
- Rename TV/anime
- CSV export

APP

- System
- Security
- Settings
- About
- Show adult content

Scan now

Problems

Things the scanner could not resolve on its own. Use **Fix...** to correct a file's details; fixes are remembered and re-applied on every rescan, so a corrected file never shows up here again.

Open problems **15**
need attention

Unparsed names **4**
no season/episode found

ffprobe errors **0**
unreadable files

Missing files **11**
seen before, gone now

Duplicate episodes **undefined**
same S/E, several files

Fixes saved **0**
0 ignored files

UNPARSED FILE NAMES (4)

LIBRARY	PATH	WHY	GUESS	
Anime	I'm a Noble on the Brink of Ruin, So I Might as Well Try Mastering Magic\I'm a Noble on the Brink of Ruin, So I Might as Well Try Mastering Magic_S01E1.mp4	could not find episode number	I'm a Noble on the Brink of Ruin, So I Might as Well Try Mastering Magic	Fix...
Anime	I'm a Noble on the Brink of Ruin, So I Might as Well Try Mastering Magic\I'm a Noble on the Brink of Ruin, So I Might as Well Try Mastering Magic_S01E1.mp4	could not find episode number	I'm a Noble on the Brink of Ruin, So I Might as Well Try Mastering Magic	Fix...
TV	Star Trek Picard\Star Trek Picard S3\10600367.mp4	could not find episode number	Star Trek Picard S3	Fix...
TV	Superhero Fight Club 2.0\Superhero Fight Club 2.0 (2016).mp4	could not find episode number	Superhero Fight Club 2 0	Fix...

FFPROBE ERRORS (0)

LIBRARY	PATH	ERROR
Nothing here.		

DUPLICATE EPISODES (UNDEFINED)

LIBRARY	SERIES	EP	FILES	PATHS
Nothing here.				

MISSING FILES (11) **Forget missing files**

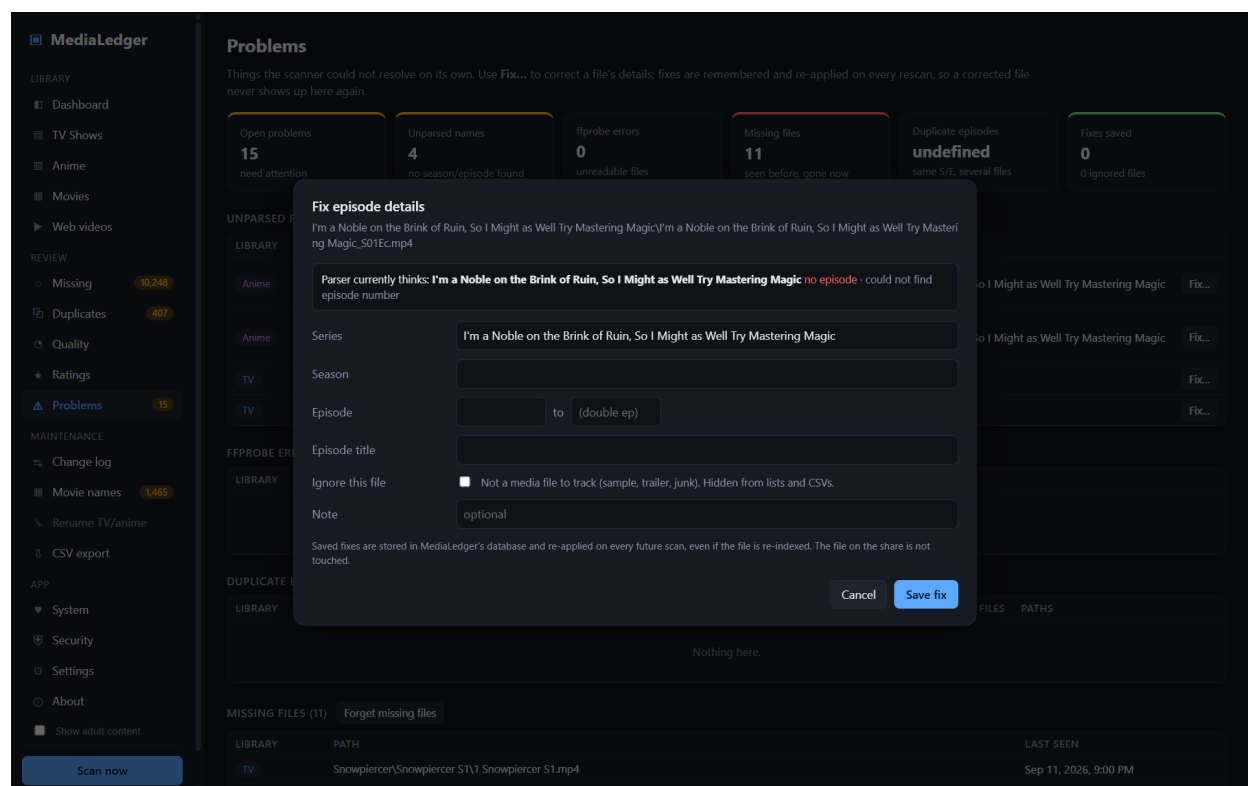
LIBRARY	PATH	LAST SEEN
TV	Snowpiercer\Snowpiercer S1\1 Snowpiercer S1.mp4	Sep 11, 2026, 9:00 PM

Everything the scanner could not resolve.

The summary strip counts open problems. Sections below list unparsed file names with the parser's best guess, ffprobe errors, duplicate episodes, missing files, and your saved manual fixes (including ignored files).

- **Forget missing files** removes the records of files that have vanished. Until you press it they stay flagged, which protects you from a NAS outage being mistaken for a deletion.
- A root that cannot be reached during a scan is logged as `root_offline` and skipped entirely; its files are never marked missing.

13.1 The Fix dialog



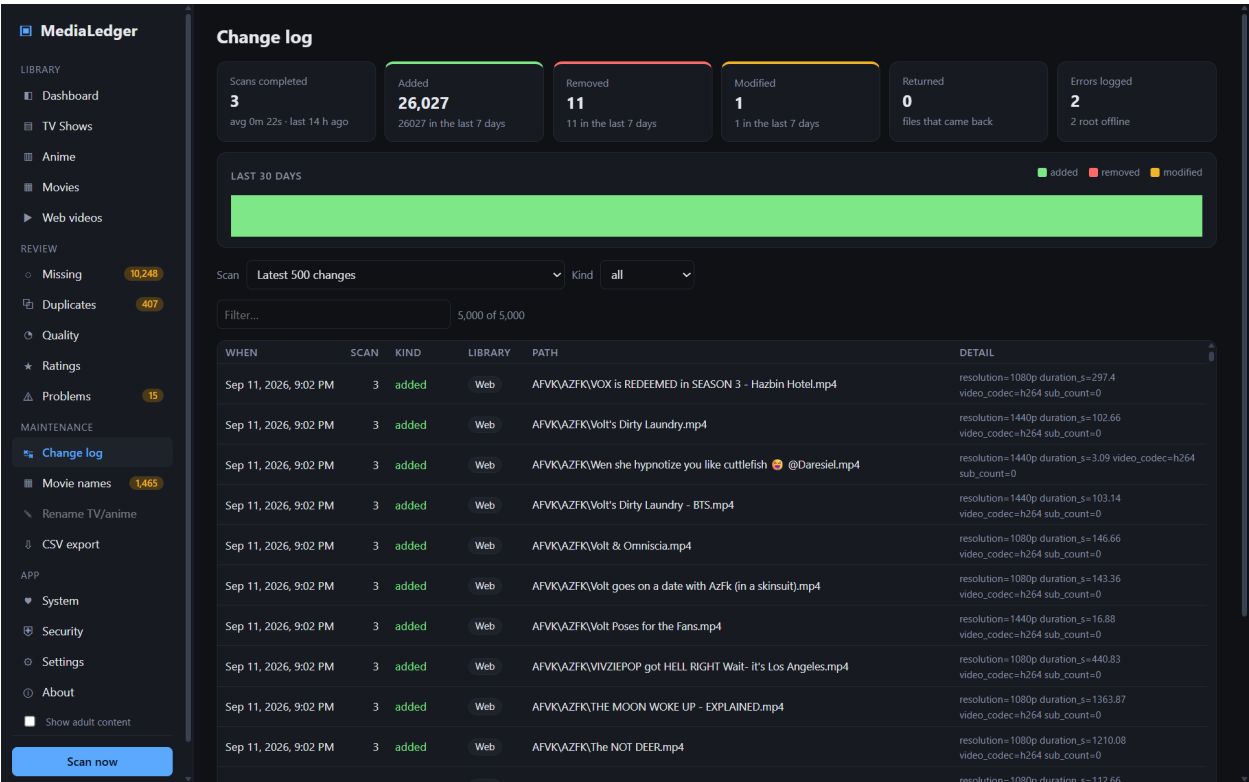
Correcting an episode that the parser could not place.

Fix... appears on every file row in the app. The dialog states what the parser currently thinks, then lets you set the correct values:

- Episodes: series (with suggestions), season, episode and an optional end episode for double episodes, episode title.
- Movies: title, year, edition tag, and **Source** (Web or Rip) for the naming engine.
- **Ignore this file** hides a file from every list, CSV and rename plan without touching it on disk.
- A free-text note.

Saved fixes are stored by root and relative path, re-applied on every future scan, and follow the file through the app's own rename tools. **Remove fix** returns a file to what the parser says.

14. Change log



The change log with its 30-day activity chart.

Every scan writes one entry per event: **added**, **removed**, **modified** (size or date changed), **returned** (a missing file came back), **probe_error**, **root_offline** and **warning**. The strip at the top counts totals and the last seven days; the chart shows activity per day for the last 30. Pick a scan in the dropdown to see only its events with its duration, files seen and thread count, or filter by kind.

15. Movie names (the naming engine)

Proposed names with flags; every batch is a dry run until the live switch is armed.

This tab renames movie files to a consistent pattern built from facts the app can prove. It is the most carefully guarded part of MediaLedger, because it is one of only two places that write to the share.

15.1 The pattern

Title (Year) - Source Resolution HDR Codec [Audio] [{edition-Name}].ext

Examples: A Breed Apart (2025) - Web 1080p SDR H264.mp4 and Avatar (2009) - Rip 4K HDR HEVC {edition-Extended Collector's Edition}.mkv.

Token	Where it comes from	When unknown
Title, Year	The parser and your manual fixes, or the Plex match when the truth source is set to Plex.	Year becomes the word Year; the file is flagged.
Source	Web when the old name carries a LiTV or WEB marker; Rip for BRrip, BluRay, BDRip, Remux or DVD; or whatever you set in the Fix dialog or with bulk source.	The word Source; the file is flagged.
Resolution	ffprobe pixel size: 4K, 1440p, 1080p, 720p, 576p, 480p or SD.	Blocked.
HDR / SDR	ffprobe colour transfer: HDR10, Dolby Vision and HLG all become HDR.	SDR.
Codec	ffprobe video codec: H264, HEVC, AV1, MPEG4, VC1...	Blocked.
Audio	ffprobe audio languages, only when they are not plain English, e.g. ENG+JPN.	Omitted; flagged if undefined.
Edition	Edition words found in the old name, in Plex's {edition-...}	Omitted.

	form.	
--	-------	--

The title keeps your existing words. Colons and question marks are dropped, slashes become dashes, other characters Windows forbids are removed, and trailing dots and spaces are trimmed. Nothing is re-capitalised.

15.2 Ready, flagged, blocked

- **Ready** – the proposed name differs from the current one and every check passes.
- **Flagged** – ready, but worth a look. Flags include `no_source`, `no_year`, `res_mismatch` (the old name claimed a resolution the probe disagrees with; the probe wins), `hdr_claimed_but_sdr`, `audio_und`, `plex_title_differs` and `plex_unlinked`. The tab explains each flag under "What the flags mean".
- **Blocked** – never renamed: no successful probe, no title, or a **collision** where two or more files would receive the identical name.

15.3 Batch settings

Setting	Effect
Layout	Rename in place changes only the file name (an atomic rename). Move into "Title (Year)" folders creates Plex's preferred folder layout by copying the file, verifying its size and a head-and-tail hash, then deleting the original.
Batch limit	The most files one live batch may contain. Forces review in chunks.
Title & year from	File name + my fixes (default) or Plex match. With Plex, unlinked files fall back to the file name and are flagged.
Allow live renames	The arming switch. Until it is on, only dry runs are possible.

15.4 Running a batch

1. Filter and tick the files you want. **Select shown** ticks every ready file currently listed.
2. Press **Dry run N**. Nothing is touched; the result window lists exactly what would happen and the batch is recorded as a dry run in the Batches table.
3. When satisfied, tick **Allow live renames**, press **Save**, then **Rename N live**.
4. A confirmation lists the renames and asks you to type **RENAME**.
5. **Pre-flight** then checks every file: it still exists, its size matches the last scan, the target name is free, the root is writable, the path is under 255 characters and no two files share a target. If any check fails, the whole batch is aborted and nothing is renamed.
6. Each rename is followed by a re-check of the file's size before the database is updated. A failure stops the batch; earlier renames stand and are journaled.

While a live batch runs, scans and the folder watcher wait.

15.5 Undo

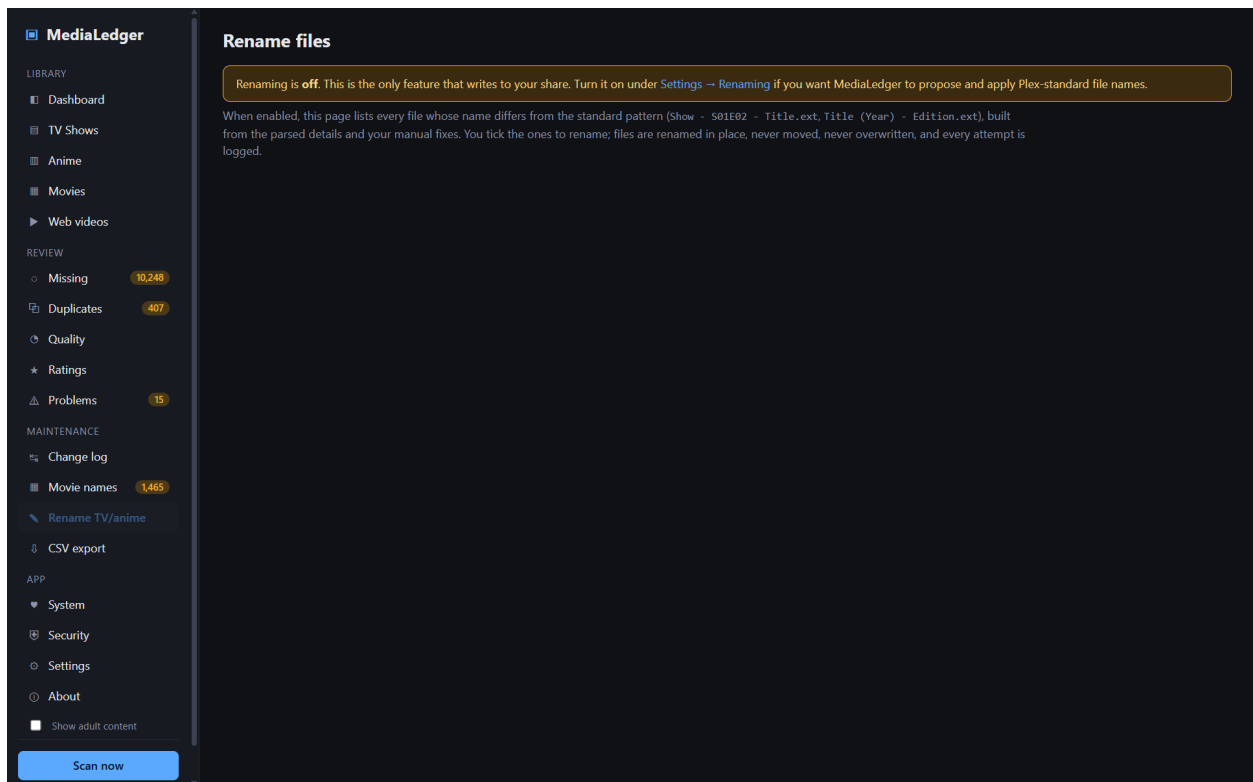
Every live batch is listed under **Batches** with an **Undo** button. Undo walks the journal in reverse; for each file it confirms the renamed file still exists with the recorded size and that the original name is free, then renames it back. Files that fail the check are left alone and reported, and the batch shows as "partially undone". **Items** shows the per-file journal of any batch.

15.6 Bulk source, collisions, placeholders

- **Set source for selected...** applies Web, Rip or "clear" to every ticked file (or to everything shown when nothing is ticked). It only records a fix; nothing is renamed.
- **Name collisions** groups the blocked pairs with their sizes. **Keep this, ignore others** marks the rest ignored (still on disk) so the kept file can be renamed; alternatively give one file a distinguishing edition via Fix....
- **Placeholders on disk** lists files already renamed with (Year) or Source in their name. Plex will not match a (Year) file, so fill the value with Fix... and rename again.

Before the first large live batch, run one small batch of 20 or 30 files in place, check the result in Explorer and in Plex, then undo it. That proves the whole path on your NAS before you trust it with the full library.

16. Rename TV / anime



The episode rename tool, shown here while disabled.

A simpler tool for episodes. It is **off** until you enable it under Settings → Renaming. When on, it lists every episode file whose name differs from Show - S01E02 - Title.ext, built from the parsed details and your fixes. Tick files, press **Rename**, confirm. Files are renamed in place, never moved or overwritten, and every attempt is logged in the History table. Fix anything wrong under Problems first, because the proposal is only as good as the parse.

17. CSV export

MediaLedger

LIBRARY

- Dashboard
- TV Shows
- Anime
- Movies
- Web videos

REVIEW

- Missing **10,248**
- Duplicates **407**
- Quality
- Ratings
- Problems **15**

MAINTENANCE

- Change log
- Movie names **1,465**
- Rename TV/anime
- CSV export**

APP

- System
- Security
- Settings
- About
- Show adult content

CSV export

Every export writes eight CSV files into a timestamped folder and refreshes the latest\ copy, so a spreadsheet can always point at the same file names.

Last export
14 h ago
9 files - 26,027 rows

Exports on record
3

After each scan
on
change in Settings

Export now Open latest folder Open exports folder C:\Project Folder\MediaLedger\devprofile\exports

FILES PRODUCED

tv_episodes.csv / anime_episodes.csv	One row per episode file with every probed field
tv_series.csv / anime_series.csv	One row per series: seasons, episodes, runtime, size, resolutions, languages, captions %, episode gaps
movies.csv	One row per movie file, with the number of versions of that title
movies_titles.csv	One row per title: file count, versions, best resolution, size
movies_multiples.csv	Only titles that have more than one file
web_videos.csv	One row per web video with channel, title, upload date and every probed field
changes.csv	The change log for the scan that triggered the export (or the latest changes)

HISTORY

WHEN	TRIGGER	SCAN	ROWS	FOLDER	
Sep 11, 2026, 10:25 PM	task	#4	26,027	C:\Project Folder\MediaLedger\devprofile\exports\20260912_0225	Open
Sep 11, 2026, 9:03 PM	task	#3	26,027	C:\Project Folder\MediaLedger\devprofile\exports\20260912_0103	Open
Sep 11, 2026, 9:01 PM	task	#2	25,266	C:\Project Folder\MediaLedger\devprofile\exports\20260912_0101	Open

Scan now

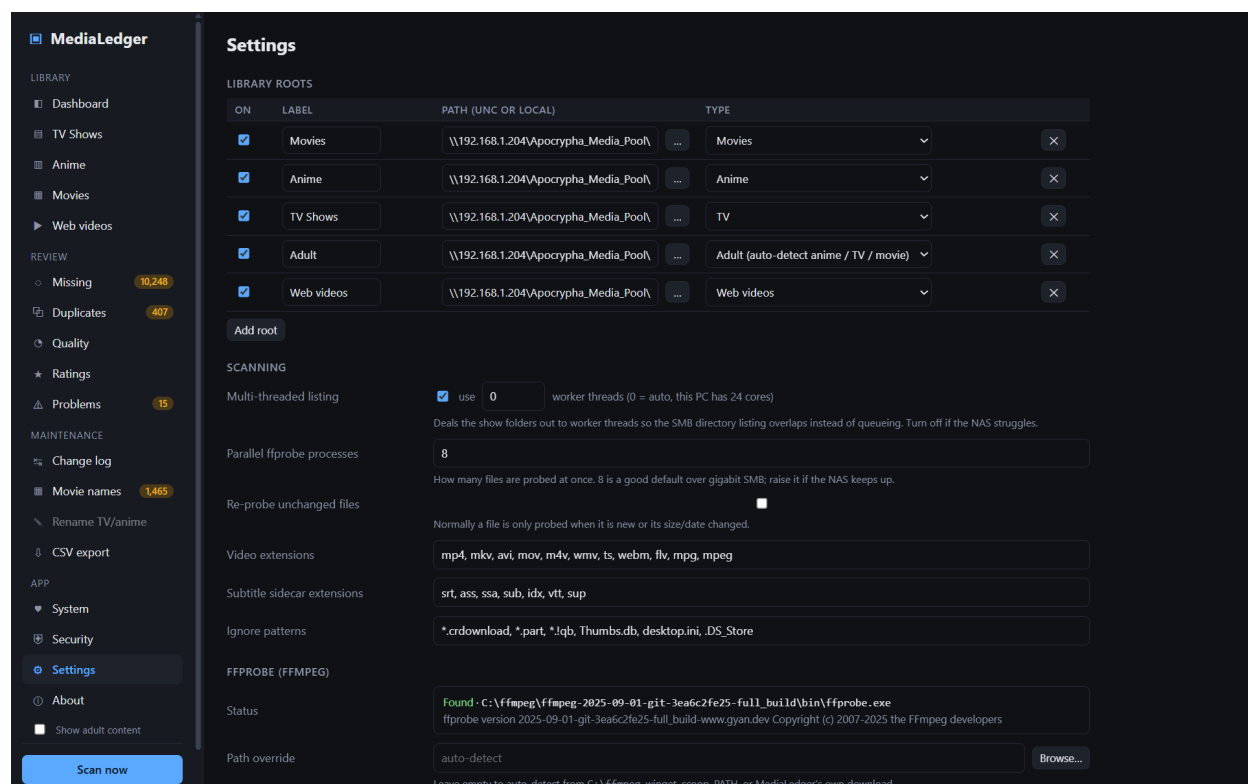
The export tab with history.

Every export writes a set of CSV files into `exports\<timestamp>\` and refreshes the `latest\` copy, so a spreadsheet can always point at the same file names. Exports run automatically after each scan (Settings → CSV export) or on demand with **Export now**. The history table lists past exports with an **Open** button.

File	Contents
tv_episodes.csv / anime_episodes.csv	One row per episode file with every parsed and probed field.
tv_series.csv / anime_series.csv	One row per series: seasons, episodes, runtime, size, resolutions, codecs, languages, captions %, episode gaps, metadata source, expected and missing episodes with the exact list, online rating, your rating and note, mixed-resolution and low-bitrate flags.
movies.csv	One row per movie file, with the number of versions of that title.
movies_titles.csv	One row per title: file count, versions, best resolution, size, languages, captions, your rating and note.
movies_multiples.csv	Only titles with more than one file.
web_videos.csv	One row per web video: channel, title, upload date, video id and every probed field.
changes.csv	The change log for the scan that triggered the export.

Adult roots are excluded unless allowed in Settings. Ignored files are always excluded.

18. Settings reference



The top of the Settings page.

Library roots

One row per folder: on/off, label, path (UNC or local; the ... button browses), and type: TV, Anime, Movies, Web videos, or Adult (auto-detect anime / TV / movie). Roots with the same type may repeat. Press **Save settings** after editing.

Scanning

- **Multi-threaded listing** deals show folders out to worker threads so directory listing over SMB overlaps. 0 threads means automatic (CPU count minus one). Turn off if the NAS struggles.
- **Parallel ffprobe processes** – how many files are probed at once; 8 suits gigabit SMB.
- **Re-probe unchanged files** – force ffprobe on everything next scan.
- **Video extensions, Subtitle sidecar extensions, Ignore patterns** – what counts as media, what counts as a caption sidecar, and glob patterns to skip.

ffprobe (ffmpeg)

Shows where ffprobe was found and its version. A path override and a **Download** button fetch the latest static build into the data folder.

Expected episodes

Turn the TVmaze / AniList lookups on or off and set how often airing series are re-checked.

Folder watch

Uses Windows change notifications on each root and starts a scan once the folder has been quiet for the settle time. A copy in progress keeps pushing the timer back.

Renaming

Enables the TV / anime rename tool. Off by default.

Adult content

Whether adult roots appear in CSVs, and the fallback subtype for files that match no pattern.

Quality thresholds

Minimum video bitrate per resolution for the Quality report.

CSV export

Output folder and whether to export after every scan.

Schedule

- **In-app timer** – scans every N hours while the window is open.
- **Windows Task Scheduler** – installs a daily task that launches MediaLedger with `--scan`, which scans, exports and exits even when the app is closed. If the app is already open, the open window runs the scan instead. Re-install the task after upgrading so it points at the current program.

Data

Database path and size, schema version, counts, and buttons to back up now, open the backups folder, the data folder and the log.

Updates

Automatic updates on/off, and the GitHub token needed only while the repository is private.

Plex

See chapter 19.

19. Plex integration

MediaLedger talks to your own Plex Media Server over its local HTTP API. It never writes to Plex and needs no Plex account: only the server's address and a token.

19.1 Setup

1. In Plex Web, open any movie or episode, click the `...` menu, choose **Get Info**, then **View XML**.
2. The page that opens has `X-Plex-Token=` at the end of its address. Copy the value after the equals sign. Do not share it; it grants access to your server.
3. In MediaLedger, Settings → Plex: enter the server URL (for Plex on the NAS, `http://192.168.1.204:32400`), paste the token, press **Test**. It reports the server version and lists the libraries it can see.
4. Press **Sync now**. The first sync derives the path mapping from the first file it recognises, for example `/media` → `\\192.168.1.204\Apocrypha_Media_Pool`, and shows it for editing.
5. Tick **Sync after every scan** to keep it current.

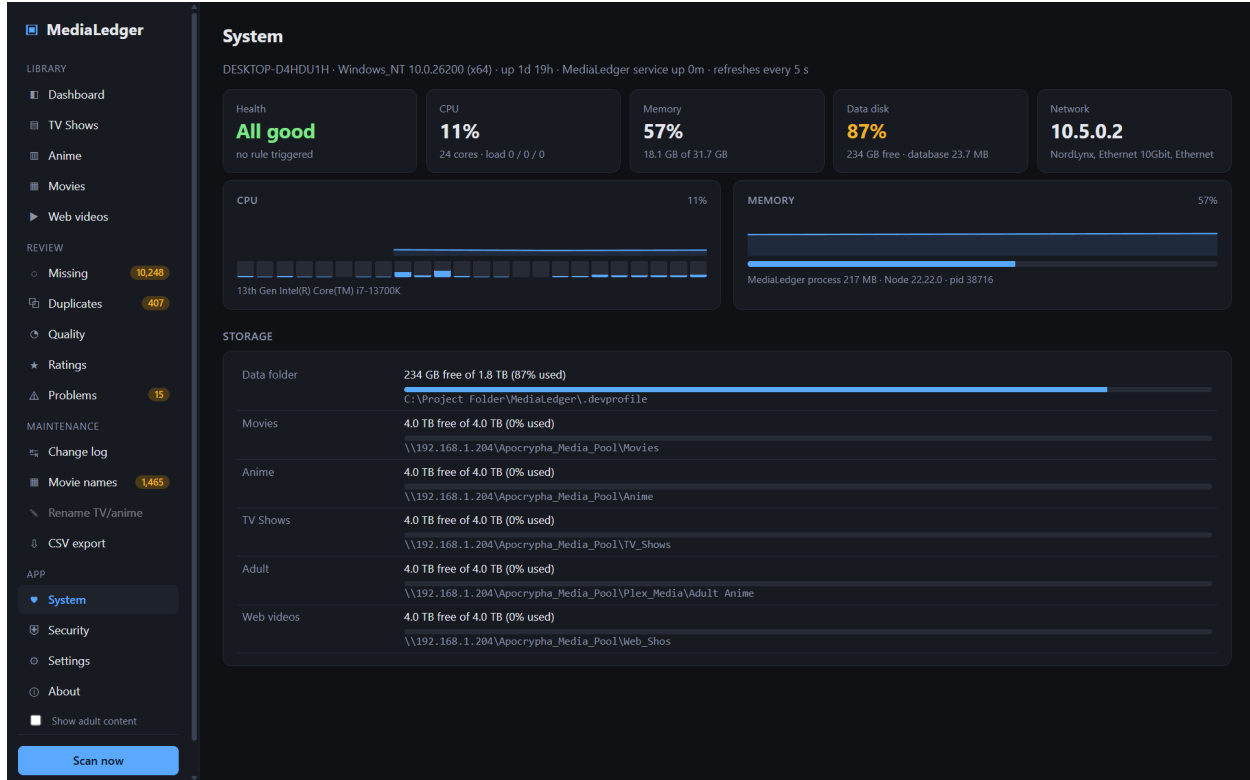
19.2 What a sync stores

Per file	Per show
Plex title and year, rating key, IMDb / TMDb / TVDB ids, your Plex rating, audience rating, play count, last viewed, resume offset, library section	Plex's title and year, ids, your rating, audience rating, episode and watched counts, content rating

These appear on the Ratings tab (Plex, Plex mine, Watched), on the series lists (Watched %, your Plex rating beside your stars), and as the optional truth source for the movie naming engine. The status line under the Sync button reports how many Plex items matched a file and lists examples of files Plex does not have.

20. System (health and hardware)

The System tab watches the machine MediaLedger runs on. On the desktop that is your PC; on the Raspberry Pi it is the board itself, where temperature and power matter. It samples every 5 seconds and keeps one hour of history in memory (nothing is written to the database or the SD card).



System tab on the desktop: health, CPU per core, memory, data disk, network, and free space on every root.

Panel	What it shows
Health	One verdict, All good, Attention or Problem, with every triggered rule listed under it. Rules: SoC over 70 °C (attention) or 80 °C (problem); throttling or under-voltage now (problem) or since boot (attention); memory over 85 / 95 %; swapping; a disk over 90 % or under 5 GB free (attention), over 97 % or under 1 GB (problem); a root that cannot be reached while no scan is running. The System entry in the sidebar shows a dot whenever the verdict is not green.
SoC temperature (Pi only)	Current chip temperature, clock speed and core voltage. The Pi firmware soft-limits at 80 °C and throttles at 85 °C; a scan on a caseless Pi 4 usually sits in the 50s to 60s.
Power & throttling (Pi only)	The firmware's throttling flags. Healthy means no under-voltage or frequency capping since boot. "Under-voltage has occurred" means the power supply or cable is weak; fix that before trusting long scans.
CPU	Overall percentage, per-core bars and the 1 / 5 / 15 minute load averages.
Memory	Used against total, swap if any, and the size of the MediaLedger process itself.
Data disk	Free space where the database, log and exports live, and the database size.
Network	Download and upload rate. During a scan the download line is the share being read.
Storage	Free space and fill percentage of the data folder and of every enabled root, measured through the share.

The thresholds live at the top of `healthOf()` in `src/main/sysmon.js` if you want to tune them for a different case or fan.

21. Security

The Security tab manages access to the **web server**. The desktop app has no login: it runs as you, on your PC, and only your PC can reach it, so on the desktop the tab simply says so. On the Raspberry Pi anyone on the LAN could otherwise open the page, and this tab is where you control that.

The screenshot shows the MediaLedger Security tab. On the left is a sidebar with navigation options: LIBRARY (Dashboard, TV Shows, Anime, Movies, Web videos), REVIEW (Missing: 10,248, Duplicates: 407, Quality, Ratings, Problems: 15), MAINTENANCE (Change log, Movie names: 1,465, Rename TV/anime, CSV export), APP (System, Security, Settings, About), and a Scan now button. The main content area is titled 'Security' and shows a status bar: 'Web server on HTTP port 8090 · 3 active sessions · 0 failed sign-ins in 24 h · 0 addresses locked out'.

Below the status bar are several security checks:

- Password set**: script-hashed in web.json (mode 0600)
- Two-factor codes**: optional: turn on below so a leaked password alone is not enough
- LAN-only access**: connections from outside private address ranges are refused
- HTTPS**: plain HTTP: fine on a trusted LAN; see the Pi guide to enable TLS
- Not running as root**: service user has no shell and no sudo
- Secrets file permissions**: web.json mode 666
- Idle sign-out**: optional: sessions last 30 days unless signed out
- Renaming switched off**: live renames need the switch on, then a password re-entry per batch

The main configuration area has two sections:

- PASSWORD**: Fields for Current, New (8+ chars), and Repeat. A 'Change password' button is present with the note 'Signs out every other session.' Below are 'OPTIONS' for LAN only (checked) and Idle sign-out (0 minutes, 0 = off). A 'Save options' button is at the bottom.
- TWO-FACTOR CODES**: A toggle set to 'off'. Description: 'Adds a code from Google Authenticator, Aegis, Bitwarden, 1Password or any TOTP app. Even a leaked password then cannot sign in.' A 'Set up 2FA' button is present.

Below these is a 'SESSIONS' table:

WHERE	BROWSER	LAST SEEN	
127.0.0.1	this (Windows NT 10.0; Win64; x64) AppleWebKit/537.36	just now	
127.0.0.1	(Windows NT 10.0; Win64; x64) AppleWebKit/537.36	just now	Sign out
127.0.0.1	curl/8.18.0	25 min ago	Sign out

Buttons: 'Sign out other sessions' and 'Sign out here'.

At the bottom is an 'AUDIT LOG' table:

WHEN	EVENT	ADDRESS	DETAIL
Sep 12, 2026, 12:52 PM	Signed in	127.0.0.1	password
Sep 12, 2026, 12:27 PM	Signed in	127.0.0.1	password
Sep 12, 2026, 12:16 PM	Signed in	127.0.0.1	password

Footer text: 'Lockout after 8 failures in 15 min · sensitive actions ask for the password again after 5 min · sessions last 30 days · full log in C:\Project Folder\MediaLedger\webprofile\security.log'

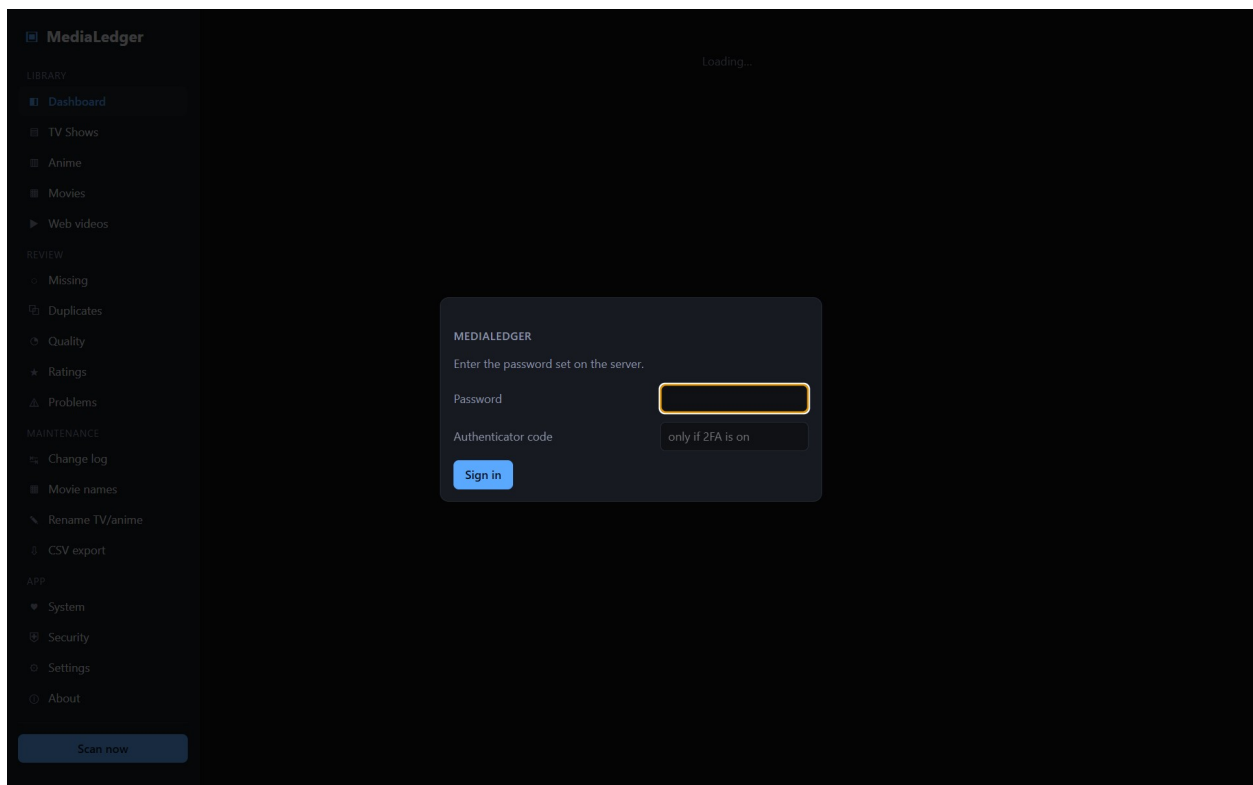
Security tab on the Pi: posture checks, password, options, two-factor codes, sessions and the audit log.

21.1 Always on

- One shared password, hashed with `crypt`, stored in `web.json` with owner-only permissions. Minimum 8 characters.
- Sessions are `HttpOnly`, `SameSite=Strict` cookies that last 30 days. Scripts cannot read them and other sites cannot use them.
- The API accepts JSON only, refuses cross-origin requests, and every page carries a strict `Content-Security-Policy` plus `X-Frame-Options: DENY`, `nosniff` and `no-referrer` headers.
- After 8 failed sign-ins in 15 minutes from one address, that address is locked out for 15 minutes.
- **LAN only**: connections from outside private address ranges are refused. Do not port-forward the Pi; use a VPN into your LAN if you need remote access.
- **Re-authentication**: a live movie rename, an undo, a TV/anime rename, purging missing files and every change on this tab ask for the password again unless you entered it within the last 5 minutes. A small dialog appears in place; the action continues once you confirm.
- Every sign-in, failure, lockout, re-authentication, sensitive action and setting change is written to `security.log` and listed at the bottom of the tab.

21.2 Controls

Control	What it does
Posture checks	Coloured tiles at the top: password set, two-factor, LAN-only, HTTPS, not running as root, secrets file permissions, share credentials permissions, idle sign-out, renaming switched off. Green is good; amber is an optional improvement; red needs attention.
Change password	Needs the current password. Signs out every other session.
LAN only	On by default. Turn off only if you know exactly which non-private network should reach the Pi.
Idle sign-out	Ends a session after N minutes without activity. 0 keeps the 30-day limit only.
Two-factor codes	Press Set up 2FA, add the shown secret (or the <code>otpauth://</code> link) to Google Authenticator, Aegis, Bitwarden, 1Password or any TOTP app, then enter the 6-digit code it shows. From then on sign-in needs password + code. Turning it off needs the password. If you lose the device, see the Pi troubleshooting table.
Sessions	Every signed-in browser with its address, browser and last activity. Sign out ends one; Sign out other sessions keeps only this one; Sign out here ends yours.
Audit log	The last 100 events with time, kind, address and detail. The full log is <code>security.log</code> in the data folder.

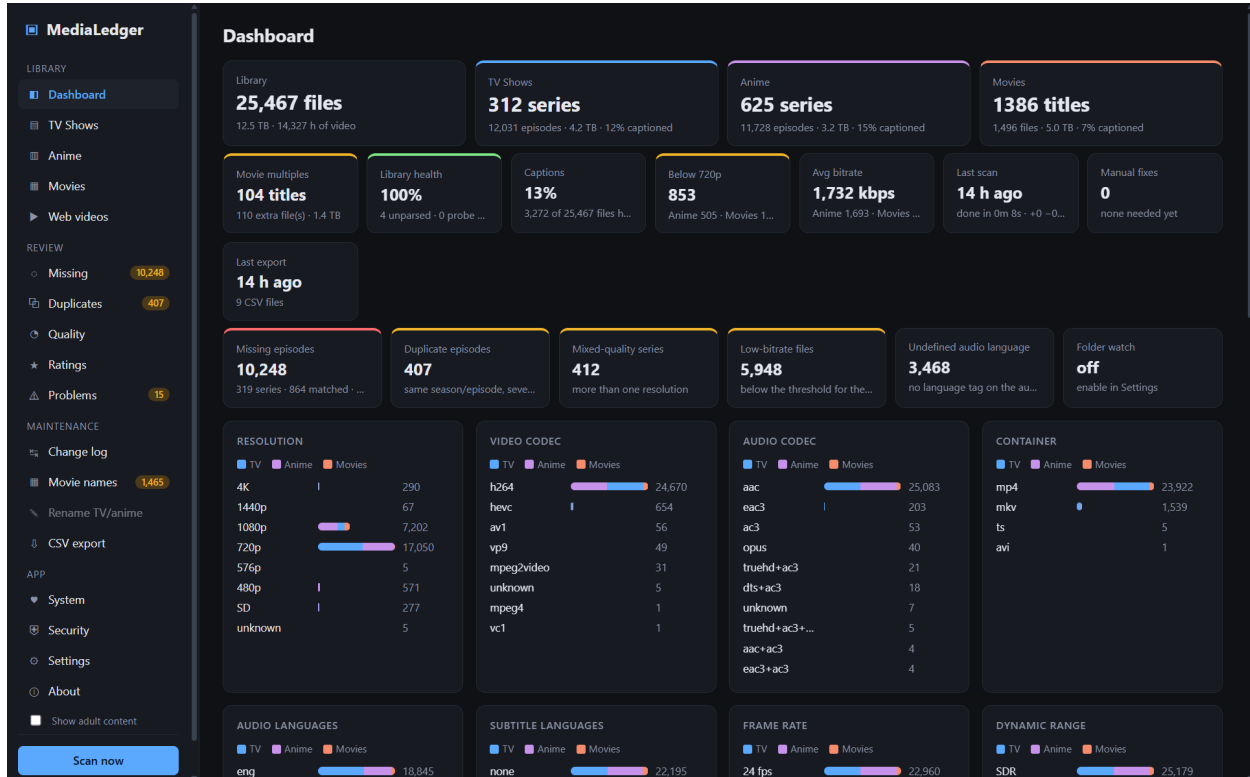


Signing in to the web server. The code field is only needed once two-factor is on.

22. Running MediaLedger on a Raspberry Pi

The same program runs as a small always-on website on a Raspberry Pi. Every tab in this manual works there, in any browser on your LAN. The Pi keeps its own database and settings; the format is identical to the desktop, so a database can be copied between the two.

Differences from the desktop: folder paths are typed instead of picked, "open folder" buttons do nothing in a browser, updates are one command on the Pi instead of the silent installer, the Windows Task Scheduler section of Settings does nothing (the in-app schedule runs because the Pi is always on), and the Security tab is live.



The dashboard served from the Pi, in a browser.

22.1 Requirements

Item	Minimum	Recommended
Board	Raspberry Pi 3 (64-bit OS)	Raspberry Pi 4B or 5, any RAM size (2 GB is plenty)
Storage	16 GB microSD, A1 class	32 GB A2 microSD or a USB SSD; the card wears under log and database writes
Network	Wi-Fi	Ethernet. Scan speed is network-bound: a Pi 4 scans at near-PC speed, a Pi 3 (100 Mbit) takes hours for the first full scan
Power	Official supply	Same; under-voltage shows on the System tab
Operating system	Raspberry Pi OS Lite (64-bit), Trixie or newer	Same. 64-bit is required: Node 22's built-in SQLite only has arm64 builds
Software	Installed by the script: Node 22, ffmpeg, cifs-utils, curl, openssl	

Access	SSH to the Pi; the NAS share username and password; a browser on the LAN	
--------	--	--

22.2 Install, step by step

1. Flash **Raspberry Pi OS Lite (64-bit)** with Raspberry Pi Imager: Choose OS → Raspberry Pi OS (other) → Raspberry Pi OS Lite (64-bit). In the customisation step set hostname `medialedger`, your user and password, time zone, and enable SSH.
2. Boot the Pi on Ethernet, then from your PC: `ssh medialedger.local`. Confirm `uname -m` prints `aarch64`.
3. Run the installer on the Pi:

```
curl -fsSL https://raw.githubusercontent.com/AxialForge/medialedger/main/server/install.sh -o install.sh && sudo bash install.sh
```

The installer is safe to rerun. It installs Node 22, ffmpeg, cifs-utils, curl and openssl; creates the `medialedger` system user (no shell, member of `video` for the hardware readings) and `/var/lib/medialedger`; downloads the **server-only release package** `medialedger-server.tar.gz` from GitHub Releases, verifies its SHA-256 and unpacks it to `/opt/medialedger`; asks once for the **NAS share username and password** and mounts `//192.168.1.204/Apocrypha_Media_Pool` at `/mnt/media` through `fstab` with the credentials in a root-only file; installs a hardened systemd service on port 8080 and the `medialedger` and `medialedger-update` commands; and asks for the **web password** (8+ characters) if none is set. At the end it prints the address.

Installer option	Effect
<code>--share=/host/share</code>	Use a different share than the default.
<code>--port=9000</code>	Serve on a different port.
<code>--https</code>	Also create a self-signed certificate; the server then serves HTTPS.
<code>--branch=main</code>	Run from the git branch instead of the release package (development).

22.3 First run on the Pi

1. Open `http://medialedger.local:8080` (or `http://<pi-address>:8080`) and sign in with the web password.
2. Open **Settings**. The roots default to `/mnt/media/Movies`, `/mnt/media/Anime` and `/mnt/media/TV_Shows`. Add `/mnt/media/Web_Shos` and `/mnt/media/Plex_Media/Adult Anime` if you use them. Save.
3. Press **Scan now**. The Pi starts with an empty ledger, so this first scan probes every file; you can close the browser and come back. Or copy the desktop database over first (22.6) and skip it.
4. Open **System** during the scan to see temperature and network load, and **Security** afterwards to set up two-factor codes if you want them.

22.4 Command reference

Command (run on the Pi)	Does
<code>sudo medialedger-update</code>	Download the newest release package, verify it, install it and restart. Data is untouched.
<code>sudo medialedger --set-password</code>	Set or reset the web password. Signs everyone out.

<code>systemctl status medialedger</code>	Is the service running, since when, last log lines.
<code>journalctl -u medialedger -f</code>	Follow the application log live.
<code>journalctl -u medialedger -n 50 --no-pager</code>	The last 50 log lines.
<code>sudo systemctl restart medialedger</code>	Restart the service (also <code>stop</code> / <code>start</code>).
<code>ls /mnt/media</code>	Is the share mounted? Should list Movies, Anime, TV_Shows and the rest.
<code>sudo mount /mnt/media</code>	Re-mount the share after a NAS reboot if the automount did not.
<code>sudo rm /etc/medialedger-cifs.cred</code> then rerun the installer	Change the share credentials.
<code>sudo bash install.sh</code>	Rerun the installer: repairs the service, mount and commands without touching data.
<code>vcgencmd measure_temp</code>	Chip temperature from the shell (the System tab shows the same).

22.5 Security on the Pi

Chapter 21 describes every control. The short version: keep **LAN only** on, never port-forward the Pi, turn on **two-factor codes** if other people use your network, and let the posture tiles guide you. For HTTPS, rerun the installer with `--https` or place `cert.pem` and `key.pem` in `/var/lib/medialedger/tls/` and restart; the browser warns once about the self-signed certificate.

The service runs as a user with no shell, with `NoNewPrivileges`, `ProtectSystem=full`, `ProtectHome` and a private `/tmp`; it can write only its data folder and the share. The share credentials are root-only in `/etc/medialedger-cifs.cred`; the web password hash, sessions and 2FA secret are owner-only in `/var/lib/medialedger/web.json`.

22.6 Moving the desktop database to the Pi

This carries every fix, keep decision, series match, rating and the whole change log across, and skips the first full scan. On the PC, close MediaLedger and copy the database to the Pi:

```
scp "%APPDATA%\MediaLedger\medialedger.db" joe@medialedger.local:/tmp/medialedger.db
```

On the Pi:

```
sudo systemctl stop medialedger && sudo mv /tmp/medialedger.db
/var/lib/medialedger/medialedger.db && sudo chown medialedger:medialedger
/var/lib/medialedger/medialedger.db && sudo systemctl start medialedger
```

Then set the roots in Settings with the same ids as on the desktop (movies, anime, tv and any you added) and the `/mnt/media/...` paths, and run a scan. Every file is recognised by its relative path and nothing is re-probed. Do not copy `settings.json`: it holds Windows paths.

22.7 File locations

Path on the Pi	Contents
<code>/opt/medialedger</code>	The application (replaced on update)
<code>/var/lib/medialedger/medialedger.db</code>	The database

<code>/var/lib/medialedger/settings.json</code>	Settings (roots, schedule, Plex, ...)
<code>/var/lib/medialedger/web.json</code>	Password hash, sessions, 2FA, options (owner-only)
<code>/var/lib/medialedger/security.log</code>	Audit log
<code>/var/lib/medialedger/medialedger.log</code>	Application log
<code>/var/lib/medialedger/exports/</code>	CSV exports
<code>/var/lib/medialedger/backups/</code>	Pre-migration database backups
<code>/var/lib/medialedger/tls/</code>	Optional cert.pem and key.pem
<code>/etc/medialedger-cifs.cred</code>	NAS share credentials (root-only)
<code>/etc/systemd/system/medialedger.service</code>	The service definition
<code>/mnt/media</code>	The share

22.8 Pi troubleshooting

Symptom	Check
Page does not load	<code>systemctl status medialedger</code> ; <code>journalctl -u medialedger -n 50</code> . Same LAN? LAN-only refuses other ranges.
Sign-in loops	Cookies blocked for the site, or the Pi clock is far off (<code>timedatectl</code>).
"Too many failed attempts"	15-minute lockout for that address; wait or restart the service.
Roots "not reachable"	<code>ls /mnt/media</code> . Empty → <code>sudo mount /mnt/media</code> ; wrong credentials → delete the cred file and rerun the installer.
Scan is slow	Ethernet, not Wi-Fi. Watch System → Network during a scan.
Renames fail: root not writable	The mount is owned by <code>medialedger</code> with write modes; check the NAS user has write permission on the share.
Forgot the web password	<code>sudo medialedger --set-password</code>
Lost the 2FA device	Stop the service, set <code>"otp": {"enabled": false}</code> in <code>/var/lib/medialedger/web.json</code> , start again.
Power & throttling tile missing	The service user must be in the video group: <code>sudo usermod -aG video medialedger</code> && <code>sudo systemctl restart medialedger</code> (the installer does this).
Above 80 °C during scans	Add a heatsink or fan, or lower Probe concurrency in Settings.

23. Troubleshooting

Symptom	Cause and fix
Scan finishes instantly and the change log says root_offline	The path in Settings is wrong or the NAS is off. Fix the path; the files are untouched and never marked missing.
Files show but resolution and length are empty	ffprobe was not found when they were scanned. Check Settings → ffprobe, download it if needed, then scan again (unprobed files are probed automatically).
A series is missing episodes it clearly has	Either the parser could not place some files (see Problems → Unparsed, use Fix...) or the online match is wrong (use Match... and pick the right entry or enter counts).
Missing shows "absolute numbering"	Episode numbers on disk continue across seasons. That season is skipped; enter per-season counts by hand or fix the episode numbers with Fix....
Movie name blocked as collision	Two files would get the same name. Keep one and ignore the other under Name collisions, or give one an edition with Fix....
Rename batch aborted in pre-flight	Something changed since the last scan (size differs, target exists, share read-only). The report names each problem; fix or deselect those files and run again. Nothing was renamed.
Plex sync matched 0 items	The path mapping is wrong. Compare a Plex file path (from View XML) with the same file's path in MediaLedger and set the mapping under Settings → Plex.
Adult tab is not there	Either no Adult root is enabled, or the Show adult content switch is off (it resets on every launch).
The app will not start a second copy	Only one instance runs; launching again focuses the open window. A scheduled --scan hands its request to the open window.

The log at %APPDATA%\MediaLedger\medialedger.log records every scan, sync, rename and error with timestamps; Settings → Data → Open log opens it.

24. Glossary

Word	Meaning
ffprobe	Part of ffmpeg; reads a media file's streams and reports resolution, codecs, languages and more without playing it.
Sidecar	A subtitle, poster or metadata file that sits next to a video with the same base name.
Override / fix	A manual correction stored in the ledger and re-applied on every scan.
Placeholder	A literal word written into a movie name where the engine could not prove a value.
Pre-flight	The set of checks run over an entire rename batch before any file is touched.
Journal	The per-file record of a rename batch that makes undo possible.
Truth source	Where the naming engine takes title and year from: file name plus fixes, or the Plex match.
Root	A watched folder with a library type.
Rating key	Plex's internal id for an item.
TOTP	Time-based one-time password: the 6-digit codes an authenticator app shows, used for two-factor sign-in on the web server.
LAN only	The web server rule that refuses connections from public address ranges.
Re-authentication	Entering the password again before an action that changes files or security settings.